

Math 220B Final

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Problem 1. Let R be a commutative ring such that $R_{\mathfrak{m}}$ is a Noetherian ring for every maximal ideal $\mathfrak{m}$. Prove or disprove that R is Noetherian.

Solution. We'll disprove the statement, by providing a counterexample.

Consider the direct product ring $R := \prod_{n \in \mathbb{N}} \mathbb{Z}/2\mathbb{Z}$ (direct product of $\mathbb{Z}/2\mathbb{Z}$ with itself over $\mathbb{N}$ as index set). We claim that this ring has the localization at all maximal ideal being Noetherian, while R itself is not. More precisely, we'll prove the following statements in order:

- (1) R is not Noetherian.
- (2) Its localization at all maximal ideal is Noetherian.

Proof of (1):

Let $I_n := \prod_{i=0}^n \mathbb{Z}/2\mathbb{Z}$ in R (i.e. every sequence in R that's of the form $(a_0, a_1, \dots, a_n, 0, \dots)$ that is 0 after index n). Notice that it's an ideal: It's clear why it's an additive subgroup, and it's an ideal because all $(a_0, a_1, \dots, a_n, 0, \dots) \in I_n$ and any $(b_n)_{n \in \mathbb{N}} \in R$ has the following:

$$(a_0, a_1, \dots, a_n, 0, \dots)(b_0, b_1, \dots, b_n, b_{n+1}, \dots) = (a_0 b_0, a_1 b_1, \dots, a_n b_n, 0, \dots) \in I_n \quad (1.1)$$

Now, since I_n allows every element with $a_k = 0$ for all $k > n$, then in particular $I_n \subset I_{n+1}$ (as all elements in I_n has $a_k = 0$ for all $k > (n+1)$), and the inclusion is strict, as the element $(0, \dots, 0, 1, 0, \dots) \in I_{n+1} \setminus I_n$ (where the 1 appears at the $(n+1)$ th entry). So, with the strict increasing chain of ideals $I_0 \subset I_1 \subset \dots \subset I_n \subset \dots$, we concluded that R is not Noetherian.

Proof of (2):

Notice that R satisfies the following: For any $a \in \mathbb{Z}/2\mathbb{Z}$, since $a^2 = a$, then any $(a_n)_{n \in \mathbb{N}} \in R$ satisfies $((a_n)_{n \in \mathbb{N}})^2 = (a_n^2)_{n \in \mathbb{N}} = (a_n)_{n \in \mathbb{N}}$. In particular, for any maximal ideal $\mathfrak{m} \subset R$, one has $R_{\mathfrak{m}}$ also satisfies this property.

Now, notice that for a commutative ring A with the property that all $a \in A$ satisfies $a^2 = a$ has all its prime ideal be maximal: Let $P \subseteq A$ denotes a prime ideal, then A/P is an integral domain. Then, for any nonzero $\bar{a} \in A/P$, we have $\bar{a}^2 = \overline{a^2} = \bar{a} = \bar{1} \cdot \bar{a}$, so $\bar{a}(\bar{a} - \bar{1}) = 0$. Which, with $\bar{a} \neq 0$, one must have $\bar{a} - \bar{1} = 0$, or $\bar{a} = \bar{1}$. This shows that A/P is $\mathbb{Z}/2\mathbb{Z}$, in particular a field (since $\{0, \bar{1}\}$ are all the elements). This concludes that P is a maximal ideal.

As a result, $R_{\mathfrak{m}}$ is a local ring with the property that all $a \in R_{\mathfrak{m}}$ satisfies $a^2 = a$, then all its prime ideals are maximal; since it only has one maximal ideal, then it only has one prime ideal $\mathfrak{m}R_{\mathfrak{m}}$. So, $\text{Nil}(R_{\mathfrak{m}}) = \mathfrak{m}R_{\mathfrak{m}}$, showing all elements in $\mathfrak{m}R_{\mathfrak{m}}$ are nilpotent. Yet, any $x \in \mathfrak{m}R_{\mathfrak{m}}$ has $x^2 = x$, so any $n \in \mathbb{N}$ satisfies $x^n = x$. Then, x is nilpotent enforces $x = 0$ (since some $n \in \mathbb{N}$ satisfies $x = x^n = 0$).

This implies that $R_{\mathfrak{m}}$ has maximal ideal 0, which must be a field, in particular it's Noetherian.

This shows a counterexample where $R_{\mathfrak{m}}$ is a Noetherian ring for all maximal ideal $\mathfrak{m}$, while R itself is not Noetherian.

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Problem 2. Let A and B be Noetherian rings together with surjective ring homomorphisms $f : A \rightarrow C$ and $g : B \rightarrow C$. Let $D = A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$ be the subring of the direct product ring $A \times B$. Prove or disprove that D is a Noetherian ring.

Solution. We'll prove that $A \times_C B$ is a Noetherian ring.

First, let $\pi_A : A \times_C B \rightarrow A$, $\pi_B : A \times_C B \rightarrow B$ be the restriction of projection from $A \times B$ onto A, B respectively. We observe that π_A, π_B are surjective:

- For any $a \in A$, consider $f(a) \in C$, by the surjectivity of $g : B \twoheadrightarrow C$, there exists $b \in B$, such that $g(b) = f(a)$, so $(a, b) \in A \times_C B$. This causes $\pi_A(a, b) = a$, showing surjectivity of π_A .
- Similarly, any $b' \in B$ with $g(b') \in C$, the surjectivity of $f : A \twoheadrightarrow C$ guarantees an $a' \in A$, such that $f(a') = g(b')$. So, $(a', b') \in A \times_C B$, $\pi_B(a', b') = b'$, showing surjectivity of π_B .

Which, we can consider the composition $\varphi = f \circ \pi_A : A \times_C B \twoheadrightarrow A \twoheadrightarrow C$, which is a surjective ring homomorphism (and it realizes C as an $A \times_C B$ -module). In particular, for all $(a, b) \in A \times_C B$ and $c \in C$, the action $(a, b) \cdot c = \varphi(a, b)c = f(a)c$.

On the other hand, also consider $\ker(\varphi) \subseteq A \times_C B$, which we claim that $\ker(\varphi) = \ker(f) \oplus \ker(g)$. For any $(a, b) \in \ker(\varphi)$, since $\varphi(a, b) = f(a) = 0$, one has $a \in \ker(f)$. On the other hand, since $g(b) = f(a) = 0$, we also have $b \in \ker(g)$, showing $(a, b) \in \ker(f) \oplus \ker(g)$, showing $\ker(\varphi) \subseteq \ker(f) \oplus \ker(g)$; on the other hand, arbitrary $a' \in \ker(f)$ and $b' \in \ker(g)$ satisfies $f(a') = 0 = g(b')$, hence $(a', b') \in A \times_C B$ and $\varphi(a', b') = f(a') = 0$, so $(a', b') \in \ker(\varphi)$, showing $\ker(f) \oplus \ker(g) \subseteq \ker(\varphi)$. This shows $\ker(\varphi) = \ker(f) \oplus \ker(g)$ as set.

Now, we'll claim the following statements:

- (1) As $A \times_C B$ -modules, A, B, C have all submodules being precisely their ideals.
- (2) As $A \times_C B$ -module, A, B, C are Noetherian. In particular, all ideals of A, B, C are Noetherian over $A \times_C B$.
- (3) As $A \times_C B$ -module, $\ker(\varphi) \cong \ker(f) \oplus \ker(g)$.

Then, we'll use these information to conclude the statement.

Proof of (1):

Here, we'll prove something that's applicable for all A, B, C :

Lemma. If $\varphi : R \twoheadrightarrow S$ is a surjective ring homomorphism, then defining S as an R -module with such homomorphism, the R -submodules of S are precisely its ideals.

Proof. For any R -submodule $I \subseteq S$, since $\varphi : R \twoheadrightarrow S$ is also an R -linear map, one has $\varphi^{-1}(I) \subseteq R$ being an R -submodule. But, the R -submodule of R are precisely its ideals, so $\varphi^{-1}(I)$ is an ideal. With surjectivity of φ , $I = \varphi(\varphi^{-1}(I))$ is an ideal of S .

Conversely, any ideal $I \subseteq S$ is of the form $I = \varphi(\varphi^{-1}(I))$, with $\varphi^{-1}(I) \subseteq R$ be an ideal (in particular, an R -submodule), I is also an R -submodule of S . $\square$

Then, since the surjective ring homomorphisms $\pi_A : A \times_C B \twoheadrightarrow A$, $\pi_B : A \times_C B \twoheadrightarrow B$, and $\varphi : A \times_C B \twoheadrightarrow C$ define each ring as $A \times_C B$ -modules respectively, all of their $A \times_C B$ -submodules are precisely their ideals.

Proof of (2):

Again, we'll prove something that's applicable for all A, B, C :

Lemma. If $\varphi : R \twoheadrightarrow S$ is a surjective ring homomorphism, then S is a Noetherian ring $\implies$ it's a Noetherian R -module.

Proof. In (1) we've proven all its R -submodules are its ideals $I \subseteq S$. Which, S is a Noetherian ring implies I is finitely generated, say $I = (a_1, \dots, a_n)$. Then, for any $a \in I$, there exists $s_1, \dots, s_n \in S$, where $a = \sum_{i=1}^n s_i a_i$; by the surjectivity of φ , there exists $r_1, \dots, r_n \in R$, such that $\varphi(r_i) = s_i$ for all index i . Hence, the following holds:

$$a = \sum_{i=1}^n s_i a_i = \sum_{i=1}^n \varphi(r_i) a_i = \sum_{i=1}^n r_i \cdot a_i \quad (2.1)$$

This shows that I as R -submodule can be generated by $a_1, \dots, a_n \in I$, in particular it's finitely generated. Hence, with all R -submodules being finitely generated, S is a Noetherian R -module. $\square$

Now, since A, B are assumed to be Noetherian rings, they're automatically Noetherian $A \times_C B$ -modules based on the lemma (since their module structures are given by surjective ring homomorphisms); also, since $f : A \twoheadrightarrow C$ is surjective, with A being Noetherian ring, C is also Noetherian ring. Hence, C is also a Noetherian $A \times_C B$ -module with the same logic.

As a result, all of their submodules / ideals are Noetherian, in particular $\ker(f) \subseteq A$, $\ker(g) \subseteq B$ are Noetherian $A \times_C B$ -modules.

Proof of (3):

It's clear that set theoretic wise $\ker(\varphi) = \ker(f) \oplus \ker(g)$ (proven at the start that within $A \times_C B$, they're the same set wise), in particular their additive group structures are also the same. Hence, it suffices to verify the $A \times_C B$ -module structures are compatible.

As a recall, for any $(a, b) \in A \times_C B$, and any $a' \in \ker(f)$, $b' \in \ker(g)$, the actions are as follow:

$$(a, b) \cdot a' = \pi_A(a, b)a' = aa' \in \ker(f), \quad (a, b) \cdot b' = \pi_B(a, b)b' = bb' \in \ker(g) \quad (2.2)$$

Hence, one has the following equality on the actions:

$$(a, b)(a', b') = (aa', bb') = (\pi_A(a, b)a', \pi_B(a, b)b') = ((a, b) \cdot a', (a, b) \cdot b') = (a, b) \cdot (a', b') \quad (2.3)$$

where the left most side is the multiplication on $\ker(\varphi) \subseteq A \times_C B$, while the right most side indicates the $A \times_C B$ -action on $\ker(f) \oplus \ker(g)$. This shows that as $A \times_C B$ -module, $\ker(\varphi) \cong \ker(f) \oplus \ker(g)$.

Finally, going back to the statement, we have the following exact sequence of $A \times_C B$ -modules:

$$0 \longrightarrow \ker(\varphi) \hookrightarrow A \times_C B \xrightarrow{\varphi} C \longrightarrow 0$$

Hence, one has $A \times_C B$ is Noetherian ring (i.e. Noetherian as module over itself) $\iff \ker(\varphi)$ and C are Noetherian $A \times_C B$ -modules. Which, we know C is already Noetherian $A \times_C B$ -module by (2), while $\ker(\varphi) \cong \ker(f) \oplus \ker(g)$ as $A \times_C B$ -module, so (2) shows that $\ker(f), \ker(g)$ are Noetherian $A \times_C B$ -module, so is $\ker(\varphi)$ as their direct sum.

This concludes that $A \times_C B$ must be a Noetherian ring.

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Problem 3. Let R be a commutative local ring with maximal ideal $\mathfrak{m}$. Assume that $\mathfrak{m}$ is principal and $\bigcap_{n \geq 1} \mathfrak{m}^n = (0)$. Prove or disprove that R is Noetherian.

Solution. We'll prove that R is Noetherian.

Given that $\mathfrak{m} = (a)$ for some $a \in R$ (since it's principal), then all of its powers $\mathfrak{m}^n = (a^n)$ for all $n \in \mathbb{N}$. Which, for all nonzero proper ideal $I \subseteq R$, there exists some $k \in \mathbb{N}$, such that $I \not\subseteq \mathfrak{m}^k$, since of $I \subseteq \mathfrak{m}^k$ for all $k \in \mathbb{N}$, then $I \subseteq \bigcap_{n \geq 1} \mathfrak{m}^n = (0)$, so I nonzero enforces such k to exists.

Also, since I is a proper ideal, then $I \subseteq \mathfrak{m}$ (all proper ideal is contained in a maximal ideal, and here we only have one choice because the ring is local), so there exists $n \in \mathbb{N}$, such that $I \subseteq \mathfrak{m}^n$.

Based on the above two facts, choose the minimum $k \in \mathbb{N}$ such that $I \subseteq \mathfrak{m}^k = (a^k)$, while $I \not\subseteq \mathfrak{m}^{k+1} = (a^{k+1})$. Then, there exists element $i \in I \setminus \mathfrak{m}^{k+1}$, which it can be expressed as $i = ra^k$ for some $r \in R$ (since $i \in \mathfrak{m}^k$), while $i \neq r'a^{k+1}$ for all $r' \in R$ (since $i \notin \mathfrak{m}^{k+1}$).

Now, we claim that r must be a unit: Suppose the contrary r is not a unit, then since in R the local ring we have every element in $R \setminus \mathfrak{m}$ being a unit, this enforces $r \in \mathfrak{m}$. Hence, there exists $r' \in R$, such that $r = r'a$. However, this shows that $i = ra^k = (r'a)a^k = r'a^{k+1} \in \mathfrak{m}^{k+1}$, contradicting our assumption $i \notin \mathfrak{m}^{k+1}$. As a result, r must be a unit, and $a^k = r^{-1}i \in I$. This shows that $\mathfrak{m}^k = (a^k) \subseteq I \subseteq \mathfrak{m}^k$, showing $I = \mathfrak{m}^k = (a^k)$. Hence, all ideal of R is a power of the maximal ideal, and is a principal ideal.

As a result, since all ideals are principal (since it's true for all nonzero proper ideals, whiler (0) and R are trivially principal), R is a Principal Ideal Ring, in particular Noetherian.

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Problem 4. Is there an integral domain R whose all maximal ideals are principal but R is not a PID? Justify your answer.

Solution.

To be honest, not sure (but some sources proposed that all $f(x) \in \mathbb{Q}[x]$ where $f(0) \in \mathbb{Z}$ can be one).

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Problem 5. Let R be an integral domain in which the set of principal ideals satisfies the ascending chain condition. Prove or disprove that R is Noetherian.

Solution. We'll disprove the statement, by providing a counterexample.

Consider k a field, and the integral domain $R = k[x_1, x_2, \dots]$ with (countably) infinite indeterminates. Notice that the ideal $I = (x_1, x_2, \dots)$ generated by all indeterminates is not finitely generated as R -module:

Proof: Suppose the contrary that I is a finitely generated R -module, then there exists $f_1, \dots, f_n \in I$, such that $I = \sum_{i=1}^n Rf_i = (f_1, \dots, f_n)$ as ideal. Since each polynomial only involves finitely many indeterminates, there are only finitely many indeterminates involved in $f_1, \dots, f_n$, say the indeterminates involved are contained in $\{x_1, \dots, x_k\}$.

Then, consider the element $x_{k+1} \in I$: By our assumption, there exists polynomials $g_1, \dots, g_n \in R$, such that $x_{k+1} = \sum_{i=1}^n g_i f_i$. Which, the indeterminates of $x_{k+1}, f_1, \dots, f_n, g_1, \dots, g_n$ are all contained in $\{x_1, \dots, x_m\}$ for some suitable $m \in \mathbb{N}$, one can realize the elements $x_{k+1}, f_1, \dots, f_n, g_1, \dots, g_n$ as elements in the subring $k[x_1, \dots, x_m] \hookrightarrow k[x_1, x_2, \dots]$, and inside this smaller subring, the equality $x_{k+1} = \sum_{i=1}^n g_i f_i$ still holds.

Now, consider the evaluation map $k[x_1, \dots, x_m] \rightarrow k$, by $x_{k+1} \mapsto 1$, and all other $x_r \mapsto 0$. Notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + \dots$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i , which is with constant term 0). Hence, with each $x_r \mapsto 0$ for $r \neq k+1$ (in particular, $r = 1, \dots, k$ all sends to 0), $f_i \mapsto 0$ also (since it's a polynomial involving only $x_1, \dots, x_k$, and with no constant term).

As a result, one has the following:

$$x_{k+1} = \sum_{i=1}^n g_i f_i \mapsto 0 \quad (5.1)$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{k+1} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R -module. □

This shows that R has non-finitely generated ideal, hence R is not Noetherian ring.

Now, we claim that R has the set of principal ideals satisfy the ascending chain condition. In particular, we'll show it's a UFD, and show that UFD satisfies ascending chain condition for principal ideals.

I. R is a UFD:

For each polynomial $f \in R$, since it only involves finitely many indeterminate, there exists some suitable large $N \in \mathbb{N}$, such that f only involves indeterminates in $\{x_1, \dots, x_N\}$. Hence, it's an element of the subring $k[x_1, \dots, x_N] \hookrightarrow R$. Since this subring is a UFD (finite indeterminates polynomial ring over a field), it has a unique factorization $f = f_1 \dots f_n$, for each irreducible polynomial $f_i \in k[x_1, \dots, x_N] \hookrightarrow R$. With each $f_i \in R$ remains irreducible (as it's irreducible in any $k[x_1, \dots, x_m]$ containing it), this shows that f admits a factorization in R (as such equality also holds in R).

Now, to show that the factorization is unique in R , suppose $f = f_1 \dots f_n = g_1 \dots g_m$ as the factorization. Choose large enough $M \in \mathbb{N}$, such that all polynomials $f, f_1, \dots, f_n, g_1, \dots, g_m$ only involves indeterminates $\{x_1, \dots, x_M\}$. Then, these elements are in the subring $k[x_1, \dots, x_M]$, and the equality still holds. However, since each f_i, g_j are irreducible in R , they're also irreducible in $k[x_1, \dots, x_M]$, showing the two factorizations must be equal by the UFD property of $k[x_1, \dots, x_M]$.

Hence, R is a UFD.

II UFD Satisfies the Principal Ideal A.C.C.:

First, given a UFD A , let's recall that for any nonzero $a, b \in A$ (say with unique factorization $a = a_1^{k_1} \dots a_n^{k_n}$ and $b = b_1^{l_1} \dots b_m^{l_m}$, each a_i, a_j and b_k, b_l are distinct irreducibles when $i \neq j, k \neq l$), then $a \mid b$ iff each a_i is an associate of some b_j , with $k_i \leq l_j$ if they're associates. In particular, if $a \mid b$ while $b \nmid a$, a is guaranteed to have less irreducible factors (if it has the same irreducible factors with same powers, then $b \mid a$, since they're now associates at each irreducible factors; so, either some b_j has no associates with irreducible factors of a , or some $k_i < l_j$ if a_i, b_j are associate irreducibles).

Let $(a_1) \subset (a_2) \subset \dots$ be an ascending chain of principal ideals in A . Let $a_1 = r_1^{k_1} \dots r_n^{k_n}$ be its unique factorization in R . Notice that since $a_1 \in (a_2)$ (or $a_2 \mid a_1$), then the irreducible factors of a_2 must be in $r_1, \dots, r_n$ (with powers at most k_i , if r_i is an irreducible factor of a_2). Inductively, if $a_i = r_{j_1}^{k_{j_1,i}} \dots r_{j_m}^{k_{j_m,i}}$ (where each $j_l \in \{1, \dots, n\}$, and $k_{j_l,i} \leq k_{j_l}$, the power of r_{j_l} for a_1), then each a_{i+1} must have irreducible factors from $r_{j_1}, \dots, r_{j_m}$, and with degree at most $k_{j_l,i}$ if r_{j_l} is an irreducible factor of a_{i+1} .

As a result, $(a_1) \subset (a_2) \subset \dots$ must stabilize at some point: If it's a strict chain of inclusion, then each a_{i+1} must necessarily have less irreducible factors than a_i . But, since the number of irreducible factors is bounded by a_1 (which only allows factors of $r_1^{k_1} \dots r_n^{k_n}$ for each a_i), then each inclusion $(a_i) \subset (a_{i+1})$ must remove at least one irreducible factor, showing a_2 only has at most $k_1 + \dots + k_n - 1$ irreducible factors, a_3 has at most $k_1 + \dots + k_n - 2$ irreducible factors, and inductively any $i \in \mathbb{N}$ provides a_i with at most $k_1 + \dots + k_n - i$ irreducible factors. Hence, for $i > k_1 + \dots + k_n$, it runs out of irreducible factors, and the chain becomes unit ideal and stabilizes, which contradicts.

So, by the unique factorization property, increasing chain of principal ideals must stabilize at some point in UFD.

Finally, because R is a UFD, for the set of principal ideals, it satisfies ascending chain condition. However, we've shown that R is not Noetherian, which provides a counterexample to the statement.